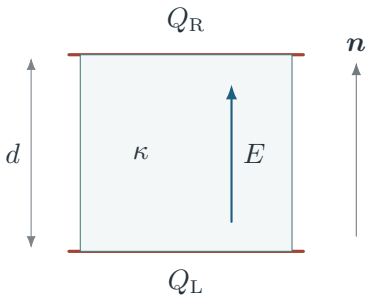
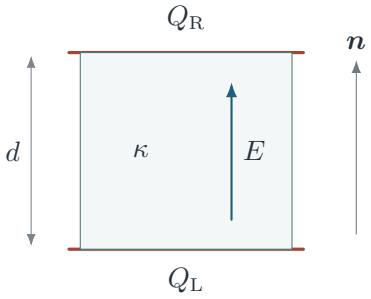
15-W1　均匀介质填满电容器



断开电源后充填

接电源时充填

图 15-8　题图　同一平行板分别采用两种电荷约束



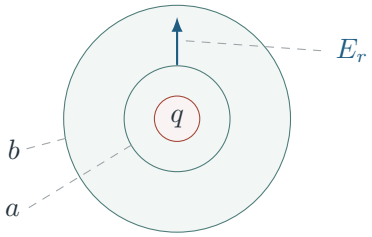
固定 Q_f：$E = E_0/\kappa$

固定 U：$E = E_0$

两种约束均有 $C = \kappa C_0$

图 15-9　解答分析图　电容由结构决定，场和储能还取决于外部连接

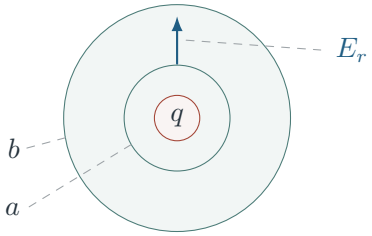
15-W2　点电荷与同心介质壳



$a < r < b$：ε

球腔与球外：ε_0

图 15-10　题图　点电荷居于球腔中心，介质只占球壳



$a < r < b$：ε

球腔与球外：ε_0

内面：$\sigma_b < 0$

外面：$\sigma_b > 0$

$\rho_b = 0$（壳内）

图 15-11　解答分析图　内表面法向朝球心，两表面的极化电荷总量抵消

15-W3　半空间介质中的导体球

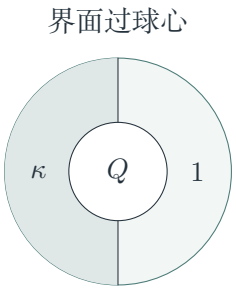
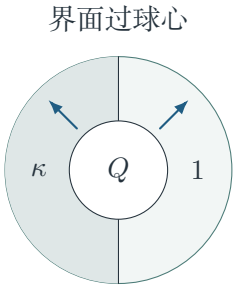


图 15-12　题图　孤立导体球处于两个半空间的交界处



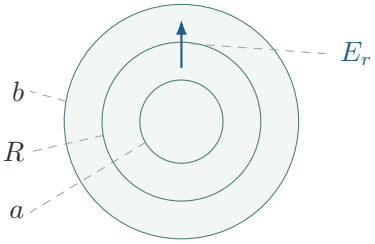
两侧 E_r 相同

$$\sigma_{f,\mathrm{d}} = \kappa\sigma_{f,\mathrm{v}}$$

$$Q_{\mathrm{d}} : Q_{\mathrm{v}} = \kappa : 1$$

图 15-13　解答分析图　验证径向场后，再按两侧介电常数积分自由电荷

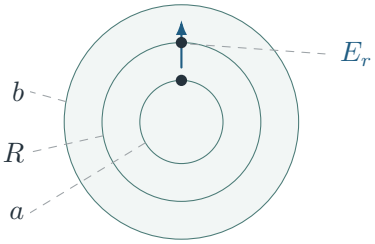
15-W4　双层球形电容器的电容和击穿



内层 $\varepsilon_1, E_{m,1}$

外层 $\varepsilon_2, E_{m,2}$

图 15-14　题图　球形电容器的两层绝缘沿径向排列



内层 $\varepsilon_1, E_{m,1}$

外层 $\varepsilon_2, E_{m,2}$

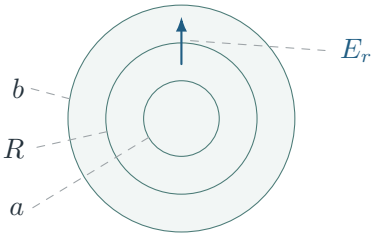
层 1 最大场：$r = a^+$

层 2 最大场：$r = R^+$

$Q_m = \min(Q_1, Q_2)$

图 15-15　解答分析图　分别检查各层内侧，最小电荷阈值决定首次击穿

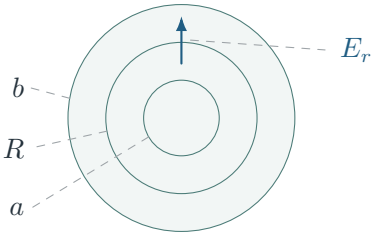
15-W5　同轴电容器的两层绝缘



同轴横截面

内极：λ

图 15-16　题图　同轴柱面的自由电荷按单位长度计算



同轴横截面

内极：λ

$E_i \propto 1/(\varepsilon_i r)$

$C' = 2\pi/\sum\int \mathrm{d}r/(\varepsilon r)$

平衡阈值含 R 的一次幂

图 15-17　解答分析图　圆柱面积含半径一次幂，不能照搬球形耐压公式

15-W6　电场跨过倾斜介质界面

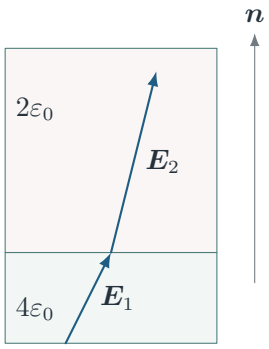
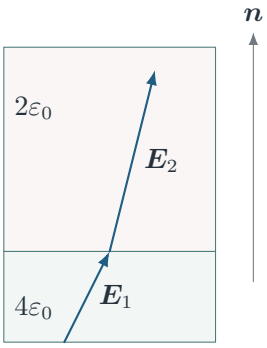


图 15-18　题图　界面法向由高介电常数侧指向低介电常数侧



$$E_{1t} = E_{2t} = E_*$$
$$E_{1n} = 2E_*,\ E_{2n} = 4E_*$$
$$\sigma_b = 2\varepsilon_0 E_*$$

图 15-19　解答分析图　两矢量采用同一比例，切向分量相同而低介电常数侧法向分量加倍

15-W7　球形气泡附近的最大场

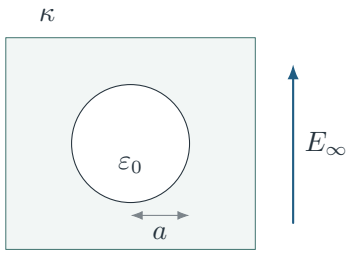
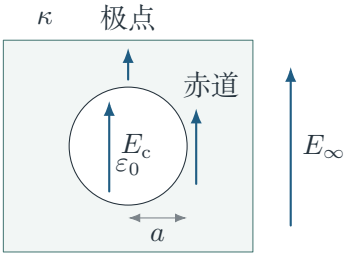


图 15-20　题图　球腔远离电极，远处介质场近似均匀



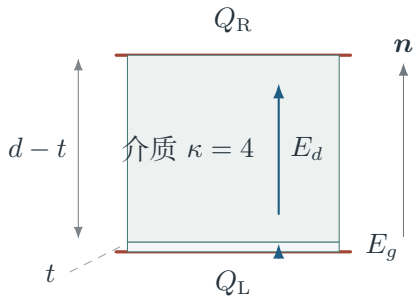
$$f = 3\kappa/(1+2\kappa)$$

介质最大场：球面赤道

腔内场：fE_∞

图 15-21　解答分析图　极点局部法向场关系不能用远处场直接替代

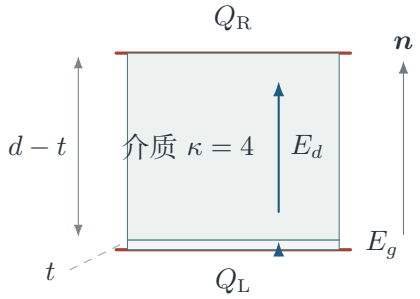
15-W8　薄真空间隙承受的电压



$t = d/20,\ \kappa = 4$

间隙跨越整个极板面积

图 15-22　题图　平面真空间隙与介质沿场方向串联



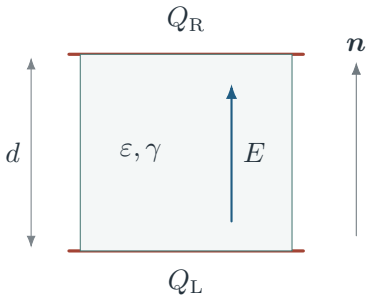
$t = d/20,\ \kappa = 4$

间隙跨越整个极板面积

$$D_g = D_d$$
$$V_g/U = 4/23$$
$$V_d/U = 19/23$$

图 15-23　解答分析图　间隙很薄仍可承担较大的电压份额

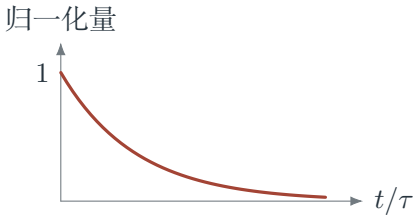
15-W9　漏电电容器的放电



与电源断开

漏电流 $j = \gamma E$

图 15-24　题图　漏电材料为孤立电容器提供内部放电通路



红线：$Q/Q_0 = U/U_0$

$\tau = \varepsilon/\gamma$

$W, H_\infty - H \propto e^{-2t/\tau}$

图 15-25　解答分析图　电荷与电压按同一时间常数衰减，场能按二倍速率衰减

15-W10　金属与介质间的空间电荷层

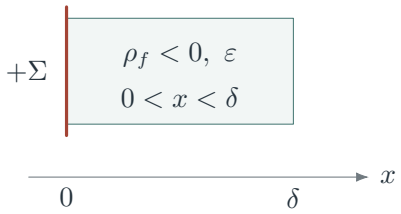
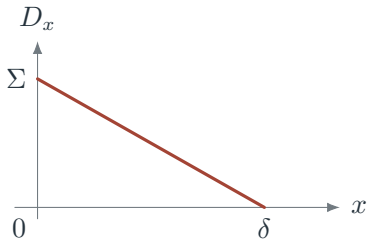


图 15-26　题图　正自由面电荷与负自由体电荷构成接触层



$$D = \Sigma(1 - x/\delta)$$
$$\Delta V = \Sigma\delta/(2\varepsilon)$$
$$W/S = \Sigma\Delta V/3$$

图 15-27　解答分析图　三角形场的面积给电势差，平方场积分给场能

15-W11　两种介质的接触电荷

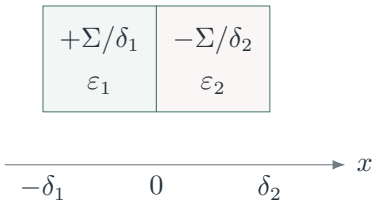
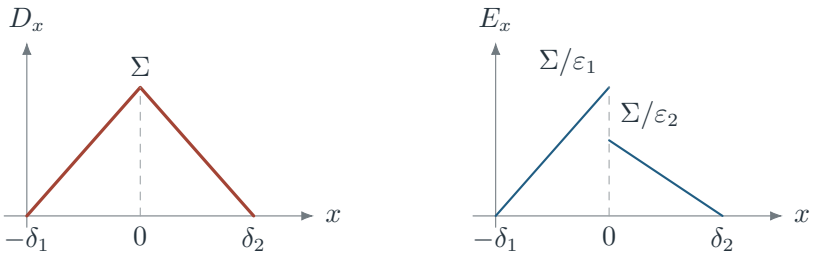


图 15-28　题图　转移电荷分别分布在两种材料的薄层中



D_n 连续，E_n 可跃变；示意 $\varepsilon_1 < \varepsilon_2$

图 15-29　解答分析图　分别绘出连续电位移与可跳变的电场，电势降由两侧电场面积相加

15-W12　接触后快速分离的功和电压

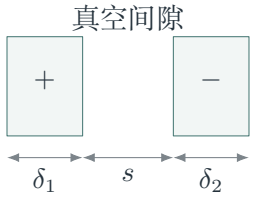
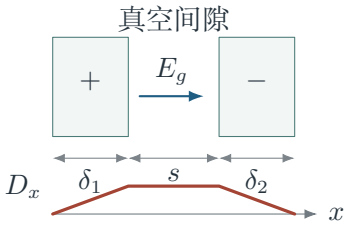


图 15-30　题图　分离形成真空间隙，两侧转移电荷暂时冻结



冻结：Σ 不变

$E_g = \Sigma/\varepsilon_0$

新增能量由机械外力提供

图 15-31　解答分析图　新增的均匀间隙场使电压和场能随距离增大

15-T1　三层平板的电压分配

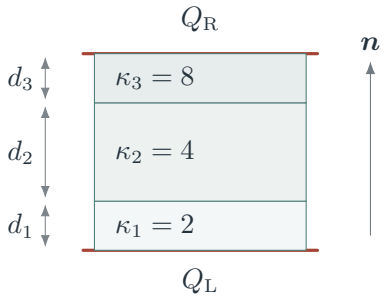


图 15-32　题图　层厚按题给数值计算，图中层厚仅作示意

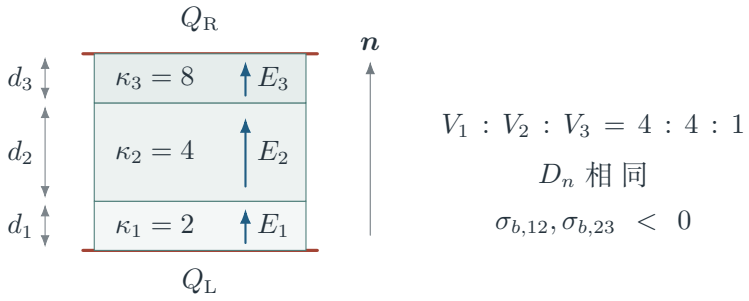
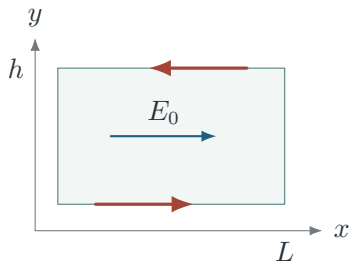


图 15-50　解答分析图　每层的电压须同时考虑场强与实际层厚

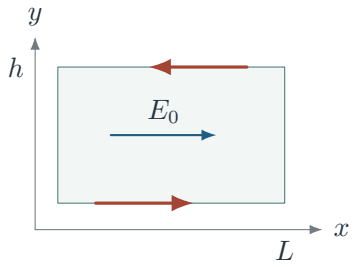
15-T2　有环流的电位移



$\varepsilon = \varepsilon_0(2 + y/h)$

绕行：逆时针

图 15-33　题图　电位移沿 x 方向，大小随 y 增大；回路按逆时针绕行



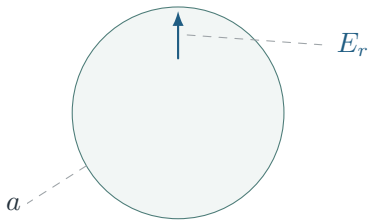
下边：$+2\varepsilon_0 E_0 L$

上边：$-3\varepsilon_0 E_0 L$

左右边：0

图 15-51　解答分析图　上边电位移更大，反向积分使总环流为负

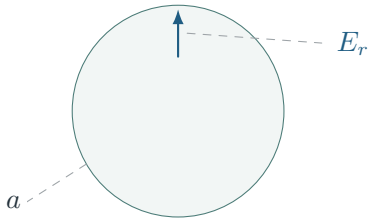
15-T3　均匀介质内的固定体电荷



球内均匀 $\rho_f > 0$

$\varepsilon = \kappa\varepsilon_0$

图 15-34　题图　自由体电荷固定，导体式电荷迁移不适用



球内均匀 $\rho_f > 0$

$\varepsilon = \kappa\varepsilon_0$

体极化电荷：负

球面极化电荷：正

球外场由总自由电荷决定

图 15-52　解答分析图　材料均匀并不保证极化体电荷为零

15-T4　锥形分区中的导体球

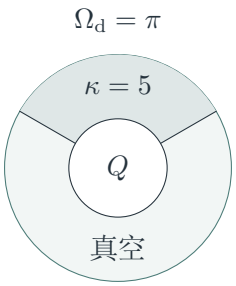
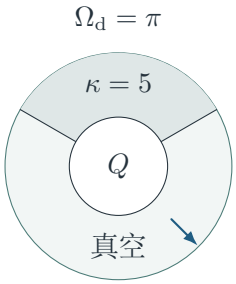


图 15-35　题图　以圆锥分区示意四分之一立体角的介质



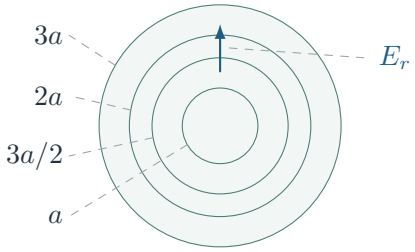
$$Q_{\mathrm{d}}/Q \;=\; 5/8$$

锥面上 $E_n \;=\; D_n \;=\; 0$

径向切向场连续

图 15-53　解答分析图　电场切于锥面，导体自由电荷向高介电常数扇区集中

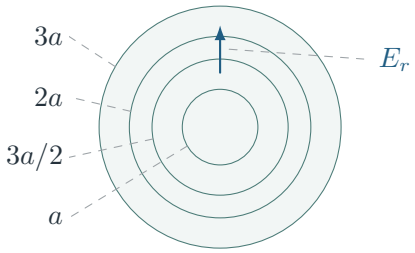
15-T5　三层球形绝缘的薄弱处



$\kappa_i \;=\; 2,5,3$

三个阈值分别给定

图 15-36　题图　各层内侧半径和阈值都参与耐压比较



$\kappa_i \;=\; 2,5,3$

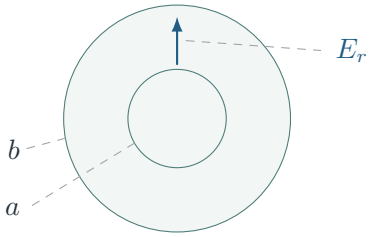
三个阈值分别给定

电荷阈值比：$6:135/8:48$

首次达到阈值：$r = a^{+}$

图 15-54　解答分析图　材料阈值最低的一层未必首先击穿

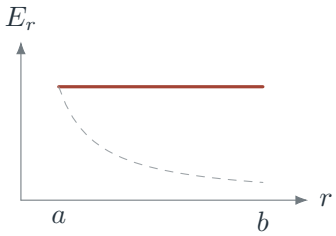
15-T6　径向渐变的同轴绝缘



同轴绝缘

$\varepsilon(r) = \varepsilon_a a/r$

图 15-37　题图　介电常数随半径减小，与圆柱电位移配合



渐变材料：常场

均匀材料：$E \propto 1/r$

$U_m = E_m(b-a)$

图 15-55　解答分析图　实线为渐变材料场，虚线为同内侧场的均匀材料

15-T7　电容与击穿电压的变化

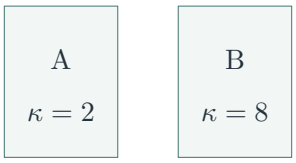
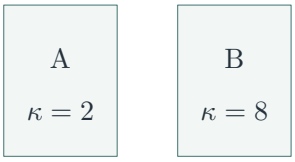


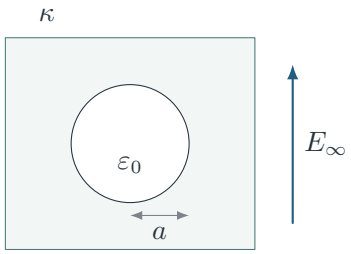
图 15-38　题图　两电容器的几何相同，材料参数分别给定



$$C_B/C_A = 4$$
$$U_{m,B}/U_{m,A} = 1/3$$
$$W_{m,B}/W_{m,A} = 4/9$$

图 15-56　解答分析图　介电常数较大仍可能对应更低的耐压和储能

15-T8　球腔两侧的场

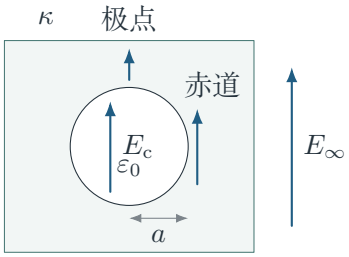


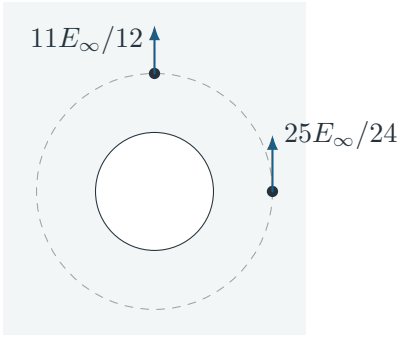
$\kappa = 4$

另考察 $r = 2a$

图 15-39　题图　球腔的极点和赤道属于不同局部场条件

$r = 2a$ 的两观察点





$\kappa = 4, \quad E_c = 4E_\infty/3$

图 15-57　解答分析图　同时画出球面两侧局部场及两倍半径处的轴线、赤道观察点

15-T9　空腔形状对局部场的影响

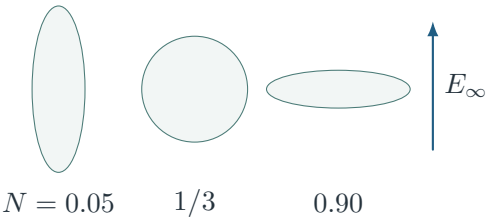


图 15-40　题图　三种示意形状的外场都沿竖直主轴

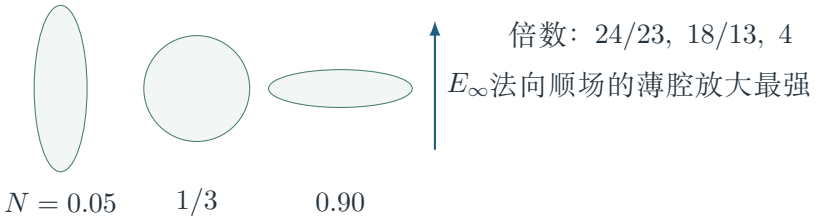


图 15-58　解答分析图　退极化因子描述形状反馈，整体尺度不改变场倍数

15-T10　带陷获面电荷的两层介质

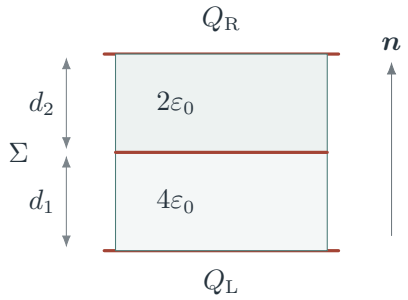
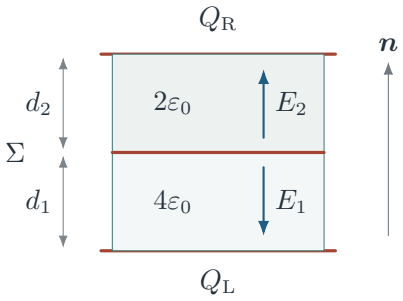


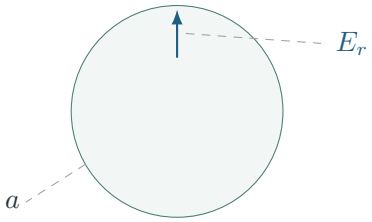
图 15-41　题图　界面有自由面电荷，电位移法向可以跳变



$$D_2 - D_1 = \Sigma$$
$$E_1 = -U/(6d)$$
$$E_2 = 13U/(6d)$$

图 15-59　解答分析图　陷获电荷使第一层场反向，两极自由电荷不再等量异号

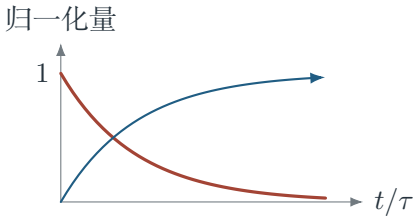
15-T11　漏电球中的体电荷迁移



球内初始 $\rho_0 > 0$

球面不允许电荷逸出

图 15-42　题图　欧姆电流把体内自由电荷送到球面



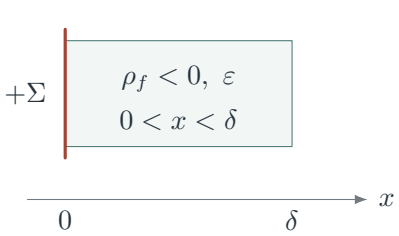
红线：ρ/ρ_0

蓝线：$3\sigma_f/(\rho_0 a)$

体电荷减少，表面电荷增加

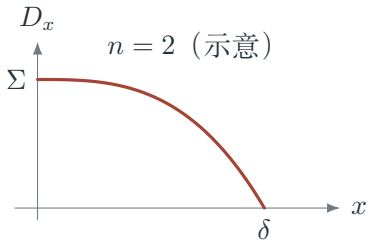
图 15-60　解答分析图　总自由电荷与球外场保持不变

15-T12　不同空间电荷剖面的电势差



$$\rho_f \propto -x^n$$
$$n \geq 0$$

图 15-43　题图　固定总电荷和厚度后，仍可改变体电荷分布



电势差：对 E 积分

场能：对 E^2 积分

电荷剖面影响两者

图 15-61　解答分析图　非均匀电荷层的电位移不再按直线下降

15-T13　帕邢曲线的最小值

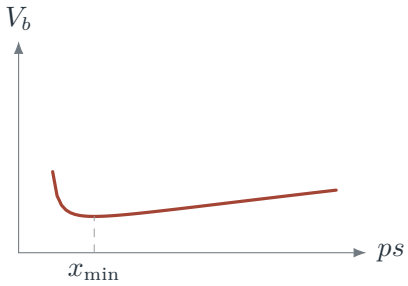
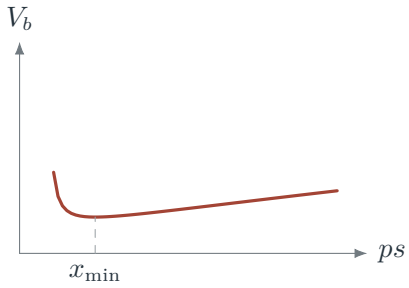


图 15-44　题图　曲线在题设模型的正分母区域内取唯一最小值



$$\ln(Ax/L_\eta) = 1$$
$$x_{\min} = \mathrm{e}L_\eta/A$$
$$V_{\min} = Bx_{\min}$$

图 15-62　解答分析图　横轴是压强与间隙的乘积，不是单独的间隙

15-T14　冻结电荷与缓慢平衡分离

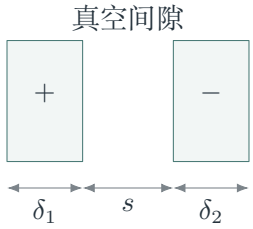
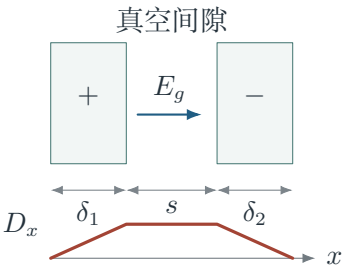


图 15-45　题图　同一分离几何分别采用两种电荷转移限制



冻结：$\Sigma = \Sigma_0$

平衡：$\Sigma = \Sigma_0/(1 + 2s/(\varepsilon_0 L))$

电荷交换改变过程约束

图 15-63　解答分析图　冻结时电压增长，维持平衡时转移电荷减少

15-T15　分离电容器接电阻后的电压峰值

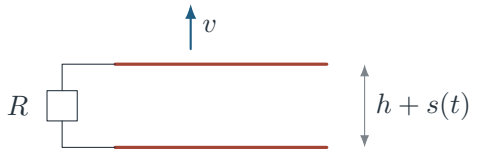


图 15-46　题图　极板分离的同时，电阻使电荷减少

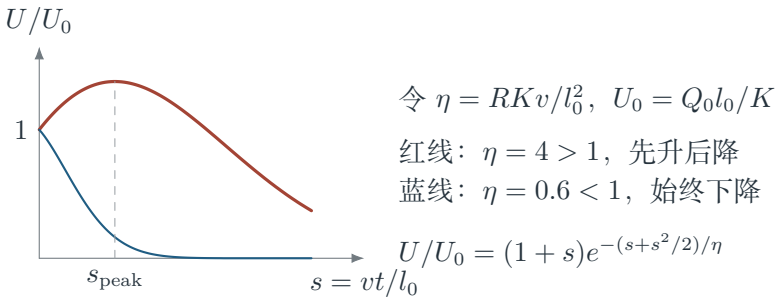


图 15-64　解答分析图　使用真实电压函数并列两类过程，峰值位置取决于分离与放电的竞争

15-T16　点电荷靠近介质平面

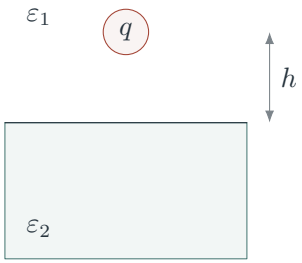
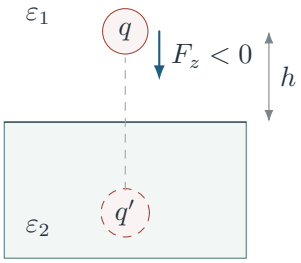


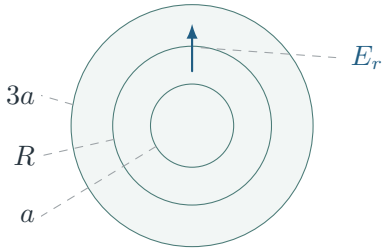
图 15-47　题图　自由电荷位于上半空间，界面无自由面电荷



图示 $\varepsilon_2 > \varepsilon_1$
反应场由像电荷表示
反应能须有因子 $1/2$

图 15-65　解答分析图　虚线像电荷表示界面响应，并非下方真实电荷

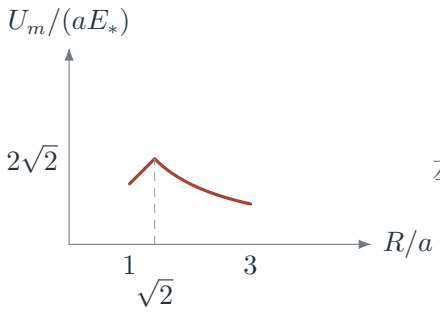
15-T17　综合题：双层绝缘的耐压设计



内层：$6\varepsilon_0$，$2E_*$

外层：$2\varepsilon_0$，$3E_*$

图 15-48　题图　界面位置可变，用两层局部阈值限制电压



$R_{\text{opt}} = \sqrt{2}a$

左段增加，右段减少

两层同时到达阈值

图 15-66　解答分析图　画出完整分段曲线，证明平衡位置确为全局最大值

15-T18　综合题：漏电引起的迟发击穿

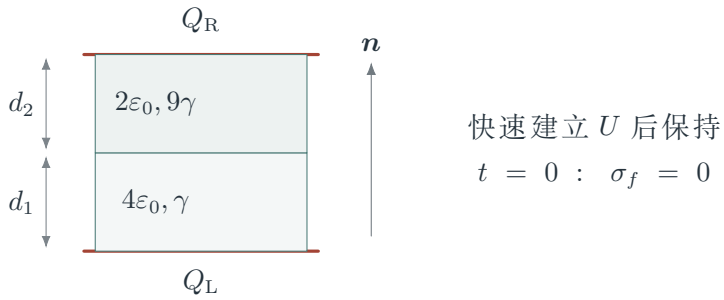


图 15-49　题图　初始按介电常数分配场，长期按电导率分配场

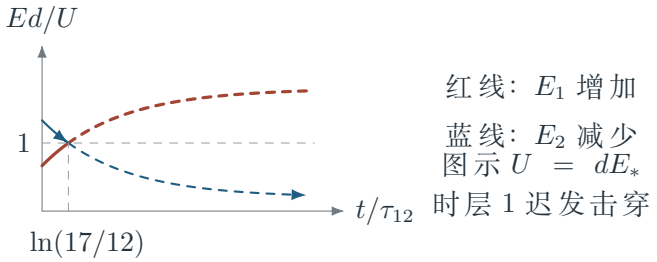


图 15-67　解答分析图　图取 $U = dE_*$；虚线为未发生击穿时的线性模型外推